

Sr. No	Question Content	Option 1	Option 2	Option 3	Option 4	Correct
1	What is a basic definition of Artificial Intelligence (AI)?	Machines that can perform any task a human can	The simulation of human intelligence processes by machines	The creation of physical robots	A type of programming language	The simulation of human intelligence processes by machines
2	Which of the following is a common application of AI?	Weather forecasting	Cooking recipes	Manual bookkeeping	Gardening	Weather forecasting
3	Which of the following techniques is primarily used for problem-solving in AI?	Data mining	Search algorithms	Web scraping	Cloud computing	Search algorithms
4	What does the term "state space" refer to in AI?	The set of all possible states in a problem	The physical space in which robots operate	A collection of databases	The network of AI agents	The set of all possible states in a problem
5	What is the primary goal of heuristic search techniques?	To explore all possible solutions	To find an optimal solution quickly	To create new heuristics	To prove all solutions exist	To find an optimal solution quickly
6	Which representation is commonly used for knowledge representation in AI?	Text files	Neural networks	Semantic networks	Spreadsheets	Semantic networks
7	What is the main difference between procedural and declarative knowledge?	Procedural knowledge is about "how," while declarative is about "what."	Procedural knowledge is stored in databases; declarative is not.	Procedural knowledge is more complex than declarative knowledge.	There is no difference.	Procedural knowledge is about "how," while declarative is about "what."
8	What does nonmonotonic reasoning allow in AI?	Static conclusions	Changing conclusions based on new information	Predicting future events	None of the above	Changing conclusions based on new information
9	What theorem is fundamental in statistical reasoning within AI?	Pythagorean Theorem	Bayes' Theorem	Fundamental Theorem of Calculus	Theorem of Relativity	Bayes' Theorem
10	What do Bayesian networks represent?	Probabilistic relationships among variables	Neural connections	Logical relations	Database schemas	Probabilistic relationships among variables
11	What is a key feature of connectionist models in AI?	They rely heavily on logic programming.	They use interconnected nodes to process information.	They are only suitable for symbolic reasoning.	They function independently of neural networks.	They use interconnected nodes to process information.
12	What does robot kinematics primarily deal with?	The emotional responses of robots	The motion of robots without considering forces	The programming of robot behaviors	The internal electronics of robots	The motion of robots without considering forces

13	Which of the following is NOT a requirement for a robot programming language?	Clarity	Efficiency	Complexity	Robustness	Complexity
14	What are production systems in AI?	Systems for manufacturing physical goods	A model consisting of rules and data structures	Platforms for deploying AI applications	Networks of AI agents	A model consisting of rules and data structures
15	Which is a design issue for search algorithms?	Cost of computation	User interface design	Networking capabilities	Hardware compatibility	Cost of computation
16	Heuristic functions are used to:	Measure the efficiency of an algorithm	Estimate the cost to reach a goal	Store knowledge in databases	Validate user input	Estimate the cost to reach a goal
17	What characterizes a constraint satisfaction problem (CSP)?	No possible solutions exist	A solution must satisfy a number of constraints	Solutions are generated randomly	They are solved using brute force only	A solution must satisfy a number of constraints
18	Logic programming is primarily associated with which type of knowledge representation?	Procedural	Declarative	Implicit	Indirect	Declarative
19	In backward reasoning, the process starts with:	Known facts and works backward to reach a conclusion	Assumptions and works forward to prove them	General principles to derive specific instances	Random facts without any order	Known facts and works backward to reach a conclusion
20	Dempster-Shafer theory is used for:	Representing uncertainty in AI	Describing robotic movements	Analyzing databases	Creating machine learning models	Representing uncertainty in AI
21	What is considered a key characteristic of intelligent agents?	They require human intervention to make decisions	They can perceive their environment and act upon it	They are only software-based	They do not need any sensory input	They can perceive their environment and act upon it
22	Which of the following is NOT a typical feature of AI systems?	Learning	Reasoning	Emotion	Problem-solving	Emotion
23	What does the Turing Test evaluate?	The speed of a computer	The ability of a machine to exhibit intelligent behavior equivalent to a human	The size of a computer's memory	The physical strength of a robot	The ability of a machine to exhibit intelligent behavior equivalent to a human
24	Which application of AI is used in medical diagnosis?	Automated transcription	Decision support systems	Weather modeling	Game playing	Decision support systems
25	In which field is AI NOT typically used?	Finance	Agriculture	Fashion design	Basic arithmetic	Basic arithmetic

26	Which AI technology is commonly used in self-driving cars?	Natural language processing	Computer vision	Database management	Blockchain	Computer vision
27	What is a common search strategy in AI?	Breadth-first search	Data mining	Static analysis	Data visualization	Breadth-first search
28	What is the purpose of using abstraction in AI?	To simplify complex problems	To increase data storage	To improve visual representation	To encrypt data	To simplify complex problems
29	A state space is defined by:	A single solution path	The initial state, actions, and goal state	Random paths without structure	The number of agents involved	The initial state, actions, and goal state
30	Which type of problem is often more complex due to multiple solutions?	Single-path problems	Multi-objective problems	Satisfiability problems	Linear problems	Multi-objective problems
31	Which technique is used to optimize search algorithms?	Random search	Hill climbing	Linear regression	Depth-first search	Hill climbing
32	What is the main goal of constraint satisfaction techniques?	Maximizing resource usage	Finding solutions that meet specific criteria	Reducing processing time	Simplifying user interfaces	Finding solutions that meet specific criteria
33	Which search method uses a priority queue?	Depth-first search	Best-first search	Linear search	Breadth-first search	Best-first search
34	Which approach uses frames to represent knowledge?	Logic programming	Semantic networks	Neural networks	Decision trees	Semantic networks
35	The 'frame problem' in AI refers to:	Difficulty in representing changing knowledge	Limitations of database storage	Issues with user interface design	The need for real-time processing	Difficulty in representing changing knowledge
36	What does a 'predicate' represent in logic?	A physical action	A condition or relationship	A data structure	A storage mechanism	A condition or relationship
37	Nonmonotonic reasoning allows for:	Static conclusions	Dynamic updates of beliefs based on new evidence	Fixed logic rules	Linear reasoning processes	Dynamic updates of beliefs based on new evidence
38	Which of the following is an example of a nonmonotonic logic?	Propositional logic	First-order logic	Default logic	Modal logic	Default logic

39	Certainty factors are used to:	Represent absolute truths	Express degrees of belief in uncertain situations	Analyze statistical data	Validate machine learning models	Express degrees of belief in uncertain situations
40	What do Bayesian networks utilize for reasoning?	Random sampling	Probabilistic inference	Deterministic algorithms	Static data analysis	Probabilistic inference
41	What is a key advantage of neural networks?	They require less data than traditional algorithms	They can model complex patterns and relationships	They are easier to understand than other models	They do not require training	They can model complex patterns and relationships
42	Recurrent neural networks are particularly suited for:	Static data	Sequential data	Image processing	Database queries	Sequential data
43	The term 'kinematics' in robotics refers to:	The control of robots	The study of motion without considering forces	The energy consumption of robots	The physical materials used in robots	The study of motion without considering forces
44	Which of the following is NOT a component of robot programming?	Control systems	Kinematic modeling	User interface design	Material selection	Material selection
45	What is a common challenge in robot programming?	Ensuring the robot is physically strong	Creating efficient algorithms for decision-making	Maintaining battery life	Designing aesthetically pleasing robots	Creating efficient algorithms for decision-making
46	A key characteristic of a production system is:	It cannot change its rules.	It consists of a set of production rules and a database.	It relies solely on neural networks.	It is always optimal in its solutions.	It consists of a set of production rules and a database.
47	Which of the following is a characteristic of a problem in AI?	Complexity	Lack of constraints	Absence of initial state	Fixed outcomes	Complexity
48	What is a major disadvantage of hill climbing?	It can get stuck in local maxima.	It requires significant memory.	It is always optimal.	It cannot be applied to complex problems.	It can get stuck in local maxima.
49	In means-ends analysis, the primary focus is on:	Finding shortcuts in problem-solving	Reducing the difference between the current state and the goal state	Establishing a sequence of actions	Analyzing the efficiency of the solution	Reducing the difference between the current state and the goal state
50	Which approach is best for representing knowledge with rules?	Semantic networks	Production systems	Neural networks	Decision trees	Production systems
51	A key feature of nonmonotonic logic is:	The preservation of truth values	The ability to withdraw conclusions	The reliance on static facts	The avoidance of uncertainty	The ability to withdraw conclusions

52	Bayesian networks can model:	Deterministic relationships only	Probabilistic dependencies	Linear equations	Non-graphical data	Probabilistic dependencies
53	The main benefit of using connectionist models is:	High interpretability	Robustness in handling noisy data	Simple design	Lower computational cost	Robustness in handling noisy data
54	What is the primary purpose of robot sensors?	To increase speed	To collect data about the environment	To minimize energy consumption	To optimize programming languages	To collect data about the environment
55	Which search algorithm is guaranteed to find a solution if one exists?	Random search	Depth-first search	Breadth-first search	Hill climbing	Breadth-first search
56	A heuristic function is designed to:	Guarantee the optimal solution	Provide a rough estimate of the cost to reach a goal	Solve problems without search	Validate the correctness of a solution	Provide a rough estimate of the cost to reach a goal
57	What is the purpose of semantic networks?	To visualize decision trees	To represent knowledge in a graph format	To store databases efficiently	To analyze statistical data	To represent knowledge in a graph format
58	Default reasoning allows us to:	Prove all conclusions are true	Make assumptions based on typical situations	Rely on fixed facts	Avoid uncertainty	Make assumptions based on typical situations
59	The concept of 'prior probability' in Bayesian inference refers to:	The probability after observing new evidence	The initial estimate of the likelihood of an event	The average of multiple outcomes	The maximum likelihood estimate	The initial estimate of the likelihood of an event
60	In a feedforward neural network:	Information only moves in one direction	There are cycles in the network	Information moves both forward and backward	It requires continuous feedback	Information only moves in one direction
61	Which of the following is an example of a narrow AI?	A self-learning robot	An AI that plays chess	A general problem solver	An AI capable of human-like conversation	An AI that plays chess
62	What is a primary goal of artificial general intelligence (AGI)?	To outperform humans in specific tasks	To understand and learn any intellectual task that a human can	To automate routine jobs	To process large datasets	To understand and learn any intellectual task that a human can
63	AI is increasingly being used in which field to enhance user experience?	Manufacturing	Customer service	Accounting	Transportation	Customer service
64	In AI, what does 'computer vision' refer to?	The ability of computers to interpret visual information	The design of computer screens	The simulation of human vision	The creation of graphic designs	The ability of computers to interpret visual information

65	In search algorithms, what is meant by 'optimality'?	The speed of the algorithm	The best possible solution in terms of cost	The simplicity of the solution	The number of solutions found	The best possible solution in terms of cost
66	Which of the following describes 'exhaustive search'?	Searching only a subset of possible solutions	Searching all possible solutions until one is found	Using heuristics to reduce search space	Prioritizing certain paths based on previous experience	Searching all possible solutions until one is found
67	What does 'solution space' refer to in AI?	The set of all possible solutions to a problem	The area where problems are defined	The environment in which a robot operates	The limits of a database	The set of all possible solutions to a problem
68	Which type of search algorithm does not guarantee finding a solution?	Depth-first search	Breadth-first search	A* search	Uniform-cost search	Depth-first search
69	What is the primary advantage of using heuristic search techniques?	They guarantee optimal solutions	They reduce computational time and resources	They can find any solution	They do not require knowledge of the problem domain	They reduce computational time and resources
70	Which of the following is an example of a heuristic function?	A random number generator	A function estimating the cost to reach the goal	A database query	A rule-based system	A function estimating the cost to reach the goal
71	In knowledge representation, what does 'ontology' refer to?	A study of human emotions	A formal representation of a set of concepts within a domain	A type of machine learning algorithm	A programming language	A formal representation of a set of concepts within a domain
72	Which representation is best for capturing hierarchical relationships?	Flat files	Semantic networks	Decision trees	Neural networks	Semantic networks
73	Which of the following best describes the concept of 'belief revision'?	Changing facts in response to new evidence	Maintaining static beliefs	Using past experiences to predict future events	Establishing rules based on historical data	Changing facts in response to new evidence
74	What is a characteristic of nonmonotonic logics?	Truth values cannot be revised	They allow for the addition of new premises without contradiction	They are purely deductive	They cannot represent uncertainty	They allow for the addition of new premises without contradiction
75	In the context of Bayesian networks, what does 'evidence' refer to?	The assumptions made during modeling	Observed data that can affect probabilities	Uncertain factors	The rules governing the network	Observed data that can affect probabilities
76	What is the purpose of Markov chains in AI?	To represent decision trees	To model random processes	To analyze neural networks	To create semantic networks	To model random processes
77	What is the role of activation functions in neural networks?	To define the architecture	To determine the output of a neuron	To optimize learning	To store knowledge	To determine the output of a neuron

78	Which type of neural network is designed to process data with a grid-like topology?	Recurrent Neural Network	Convolutional Neural Network	Feedforward Neural Network	Hopfield Network	Convolutional Neural Network
79	Which of the following is a common type of robot?	Collaborative robots (cobots)	Static robots	Digital robots	Abstract robots	Collaborative robots (cobots)
80	The term 'robot actuation' refers to:	The programming of robots	The means by which a robot moves	The sensors used in robots	The design of robot interfaces	The means by which a robot moves
81	In production systems, what are "rules" typically composed of?	A) Conditions and actions	A) Variables and constants	A) Inputs and outputs	A) Functions and algorithms	Conditions and actions
82	Which characteristic is true for production systems?	They are always complete	A) They can handle dynamic environments.	A) They require a well-defined initial state.	A) They do not rely on heuristics.	They require a well-defined initial state.
83	Which of the following is a potential drawback of using best-first search?	It guarantees optimality.	It may consume a lot of memory.	It is difficult to implement.	It ignores heuristic information.	It may consume a lot of memory.
84	Which heuristic search method is based on evaluating both cost and heuristic estimates?	Breadth-first search	Depth-first search	A* search	Greedy search	A* search
85	What does "first-order logic" enable in knowledge representation?	Only simple statements	The representation of relationships among objects	Probabilistic reasoning	Static knowledge only	The representation of relationships among objects.
86	In nonmonotonic reasoning, conclusions can be:	Always fixed	Withdrawn based on new evidence	Definitive	Only based on past experiences	Withdrawn based on new evidence.
87	What is the concept of "posterior probability" in Bayesian analysis?	The probability before observing evidence	The probability after observing evidence	The average probability of events	The maximum likelihood estimate	The probability after observing evidence.
88	What is "overfitting" in the context of neural networks?	When the model is too simple	When the model learns noise instead of the underlying pattern	When the model is generalizable to new data	When the model has high bias	When the model learns noise instead of the underlying pattern.
89	What is the role of a robot's "end effector"?	To process data	To move the robot	To interact with the environment	To control the robot's sensors	To interact with the environment.
90	What is a "goal test" in the context of AI search algorithms?	A measure of speed	A check to determine if the current state matches the goal	A performance metric for algorithms	A data validation process	A check to determine if the current state matches the goal.

91	In the context of hill climbing, what is a "local maximum"?	The overall best solution	A solution that is better than its neighbors but not the best overall	The worst possible solution	A solution with no neighbors	A solution that is better than its neighbors but not the best overall.
92	Which technique is often used to enhance the performance of genetic algorithms?	Mutation	Random sampling	Data partitioning	Heuristic evaluation	Mutation.
93	What is a "rule-based system"?	A type of unsupervised learning model	A system that uses rules to derive conclusions	A method of data storage	A programming language for robots	A system that uses rules to derive conclusions.
94	Which of the following represents "declarative knowledge"?	Knowing how to ride a bike	Knowing the capital of a country	Knowing how to swim	Knowing how to play a musical instrument	Knowing the capital of a country.
95	The principle of "closure" in nonmonotonic reasoning means:	All conclusions are final.	New evidence cannot change existing beliefs.	The addition of new information may lead to retraction of previous conclusions.	Knowledge is always static.	The addition of new information may lead to retraction of previous conclusions.
96	What does "Bayes' theorem" provide?	A method for rule-based reasoning	A way to calculate conditional probabilities	A framework for symbolic reasoning	A technique for data visualization	A way to calculate conditional probabilities.
97	What is a common use case for recurrent neural networks (RNNs)?	Image classification	Time series prediction	Static data analysis	Linear regression	Time series prediction.
98	What is "teleoperation" in robotics?	Autonomous navigation	Remote control of robots	Self-learning capabilities	Sensor integration	Remote control of robots.
99	The term "supervised learning" refers to:	Learning without labeled data	Learning from labeled data	Learning from user interactions	Learning in an unsupervised manner	Learning from labeled data.
100	What is the main difference between "classification" and "regression" in machine learning?	Classification predicts categories; regression predicts continuous values.	Classification is for supervised learning; regression is for unsupervised learning.	Classification uses neural networks; regression uses decision trees.	There is no difference; they are the same.	Classification predicts categories; regression predicts continuous values.
101	Which algorithm is primarily used for training support vector machines (SVM)?	Gradient descent	Linear regression	Quadratic programming	Random forests	Quadratic programming.
102	In reinforcement learning, what is the "reward signal"?	A penalty for wrong actions	A feedback mechanism to indicate success or failure	A learning rate adjustment	A state representation	A feedback mechanism to indicate success or failure.
103	Which of the following is a major ethical concern in AI?	Efficiency of algorithms	Data privacy and security	Cost of implementation	Speed of computation	Data privacy and security.

104	What is "algorithmic bias"?	When algorithms process data too slowly	When algorithms favor certain outcomes over others	When algorithms require too much data	When algorithms are not explainable	When algorithms favor certain outcomes over others.
105	What is a potential future trend in AI?	Decreased reliance on data	Increased collaboration between AI and humans	Elimination of all manual jobs	Reduced complexity of algorithms	Increased collaboration between AI and humans
106	What is a primary function of natural language processing (NLP) in AI?	To analyze numerical data	To enable computers to understand and generate human language	To enhance visual recognition	To manage database systems	To enable computers to understand and generate human language
107	Which of the following is a common challenge in AI development?	Limited computational power	Abundance of data	Universal applicability	Predictability of outcomes	Limited computational power
108	In which domain is AI used for fraud detection?	Education	Finance	Agriculture	Entertainment	Finance
109	What technology is primarily used in virtual assistants like Siri and Alexa?	Machine learning	Blockchain	Database management	Quantum computing	Machine learning
110	Which algorithm is often used for classification tasks in supervised learning?	K-means clustering	Decision trees	Genetic algorithms	Principal component analysis	Decision trees
111	In AI, what does "feature extraction" refer to?	The process of collecting raw data	The process of identifying relevant attributes in the data	The analysis of algorithms	The visualization of data	The process of identifying relevant attributes in the data
112	A "goal state" in AI problem-solving is defined as:	Any state in the search space	The initial condition of the problem	The desired outcome of the problem-solving process	A state that is impossible to achieve	The desired outcome of the problem-solving process
113	Which search strategy expands the shallowest node first?	Breadth-first search	Depth-first search	A* search	Hill climbing	Breadth-first search
114	What is "greedy search"?	A search that always expands the least costly node	A search that expands the node with the lowest heuristic value	A search that chooses the best immediate option without considering future consequences	A search that examines all possible paths	A search that chooses the best immediate option without considering future consequences
115	In constraint satisfaction problems, what is a "backtracking algorithm"?	An algorithm that records previous states	An algorithm that revisits previous states to find solutions	An algorithm that only searches forward	An algorithm that minimizes resource usage	An algorithm that revisits previous states to find solutions
116	What type of knowledge does "procedural knowledge" represent?	Knowledge of facts and information	Knowledge of how to perform tasks	Knowledge that is abstract and theoretical	Knowledge that is learned from observation	Knowledge of how to perform tasks

117	Which of the following best describes "semantic web"?	A web of interconnected databases	A web that allows data to be shared and reused across applications	A web of purely visual data	A web that operates without human input	A web that allows data to be shared and reused across applications
118	What is "abductive reasoning"?	Reasoning based on general rules	Reasoning that seeks the best explanation for a set of observations	Reasoning that follows a strict logical deduction	Reasoning that only considers past experiences	Reasoning that seeks the best explanation for a set of observations
119	In the context of AI, what does "entailment" refer to?	The process of drawing conclusions based on premises	The exclusion of contradictory information	The collection of data	The verification of facts	The process of drawing conclusions based on premises
120	In statistics, what is a "confounding variable"?	A variable that is dependent on the outcome	A variable that affects both the independent and dependent variables	A variable that is controlled in experiments	A variable that has no impact on the results	A variable that affects both the independent and dependent variables
121	What does "sampling" mean in the context of statistics?	The process of collecting all data	The process of selecting a subset of individuals from a population	The analysis of entire datasets	The combination of multiple datasets	The process of selecting a subset of individuals from a population
122	What is "dropout" in neural networks?	A method to reduce overfitting by randomly omitting neurons during training	A process for selecting features	A technique for initializing weights	A way to visualize data	A method to reduce overfitting by randomly omitting neurons during training
123	What type of data is typically used to train convolutional neural networks (CNNs)?	Time-series data	Text data	Image data	Graph data	Image data
124	What is the purpose of "path planning" in robotics?	To control the robot's speed	To determine a route for the robot to follow	To analyze the robot's sensors	To program the robot's responses	To determine a route for the robot to follow
125	Which of the following best describes "simultaneous localization and mapping" (SLAM)?	A method for building static maps	A technique for mapping an unknown environment while keeping track of the robot's location	A technique for data storage	A process for image recognition	A technique for mapping an unknown environment while keeping track of the robot's location
126	In a production system, what is a "state space"?	A model of all possible states the system can be in	The physical location of the robot	The storage of all rules	The algorithm used for searching	A model of all possible states the system can be in
127	What is the "initial state" in problem-solving?	The state after the goal is achieved	The starting point of the problem	The final state of the solution	The state with the highest cost	The starting point of the problem
128	Which heuristic technique uses a fixed rule to determine the next step?	Best-first search	A* search	Greedy search	Depth-first search	Greedy search
129	In hill climbing, what happens if the algorithm reaches a plateau?	It stops searching	It continues to explore other nodes	It reverts to the previous state	It guarantees a solution	It continues to explore other nodes

130	Which of the following is a disadvantage of semantic networks?	They can represent relationships effectively	They are difficult to understand	They do not handle uncertainty well	They require extensive data	They do not handle uncertainty well
131	What is the main purpose of default logic?	To create fixed rules for reasoning	To allow reasoning in the presence of incomplete information	To enforce strict logical conclusions	To model probabilistic events	To allow reasoning in the presence of incomplete information.
132	What is "multicollinearity" in regression analysis?	The presence of multiple independent variables	When two or more independent variables are highly correlated	The relationship between dependent and independent variables	The number of observations in the dataset	When two or more independent variables are highly correlated.
133	In which scenario are "generative adversarial networks" (GANs) commonly used?	Image classification	Generating new data samples	Reinforcement learning	Time series forecasting	Generating new data samples.
134	What does "robot perception" involve?	Analyzing performance metrics	Interpreting data from sensors to understand the environment	Programming robot tasks	Designing robot interfaces	Interpreting data from sensors to understand the environment.
135	What is a "smart contract" in blockchain technology?	A traditional legal contract	An agreement executed automatically by code	A paper-based agreement	A visual representation of a contract	An agreement executed automatically by code.
136	Which area uses AI for real-time language translation?	Computer vision	Natural language processing	Data mining	Predictive modeling	Natural language processing.
137	What is "autonomous AI"?	AI that operates without human intervention	AI that requires constant human supervision	AI designed for specific tasks only	AI that functions only in controlled environments	AI that operates without human intervention.
138	What is a potential application of AI in climate change?	Monitoring and analyzing environmental data	Reducing human jobs in environmental sectors	Enhancing traditional farming techniques	Increasing energy consumption	Monitoring and analyzing environmental data.
139	What does "transfer learning" help achieve in deep learning?	Faster training from scratch	Reusing a pre-trained model on a new but related task	Ignoring data quality issues	Reducing the need for labeled data	Reusing a pre-trained model on a new but related task.
140	Which algorithm is widely used in natural language processing for generating text?	Linear regression	Recurrent Neural Networks (RNNs)	Decision Trees	Support Vector Machines	Recurrent Neural Networks (RNNs).
141	What is "robot navigation"?	Designing robot hardware	The process of determining a robot's path from one point to another	Programming a robot’s tasks	Gathering sensor data	The process of determining a robot's path from one point to another.
142	What does "swarm robotics" involve?	A single robot performing tasks	A group of robots coordinating to accomplish tasks	Isolated robot systems	Manual control of multiple robots	A group of robots coordinating to accomplish tasks.

143	What is the purpose of "outlier detection" in data analysis?	To reduce data size	To identify unusual data points that differ significantly from the rest	To enhance data accuracy	To visualize data trends	To identify unusual data points that differ significantly from the rest.
144	What does "confidence level" refer to in statistics?	The degree of certainty in a statistical estimate	The average of a sample	The total number of samples collected	The probability of collecting new data	The degree of certainty in a statistical estimate.
145	What is "support vector machine" (SVM) primarily used for?	Regression analysis	Classification tasks	Clustering	Dimensionality reduction	Classification tasks.
146	What does "cross-entropy loss" measure in classification problems?	The difference between predicted and actual probabilities	The total number of samples	The speed of the algorithm	The number of layers in a neural network	The difference between predicted and actual probabilities.
147	In which industry is AI used for predictive maintenance?	Retail	Manufacturing	Education	Hospitality	Manufacturing.
148	How is AI applied in agriculture?	Manual crop monitoring	Autonomous vehicles for planting and harvesting	Basic data entry	Traditional farming methods	Autonomous vehicles for planting and harvesting.
149	What is "AI explainability"?	Making AI decisions faster	The ability to understand and interpret how AI systems make decisions	Reducing the complexity of algorithms	Enhancing the performance of AI systems	The ability to understand and interpret how AI systems make decisions.
150	What is a common challenge in developing AI systems?	Excessive data storage	Ensuring data privacy and security	Reducing computational power	Simplifying algorithms	Ensuring data privacy and security.
151	What is "zero-shot learning"?	Learning with no training data	Learning to recognize objects without prior examples	Learning only from negative examples	Learning through reinforcement	Learning to recognize objects without prior examples.
152	In the context of AI, what is a "token"?	A data point used for training	A representation of a word or term in text processing	A variable in programming	A measurement unit for data	A representation of a word or term in text processing.
153	What is the role of AI in enhancing cybersecurity?	Automating all security tasks	Identifying and responding to threats in real-time	Reducing the need for human analysts	Ignoring potential vulnerabilities	Identifying and responding to threats in real-time.
154	What is "augmented intelligence"?	The idea that AI will fully replace human intelligence	AI systems that assist and enhance human decision-making	AI functioning independently of human input	Simplified algorithms	AI systems that assist and enhance human decision-making.
155	What is "ensemble learning"?	Using multiple models to improve prediction accuracy	Training a single model repeatedly	Reducing the number of features	Ignoring outliers in data	Using multiple models to improve prediction accuracy.

156	What does "data labeling" involve?	Organizing data into folders	Assigning tags or categories to data for supervised learning	Ignoring unimportant data	Deleting redundant data	Assigning tags or categories to data for supervised learning.
157	What is "human-robot interaction" (HRI)?	The study of robots in isolation	The field focusing on how humans and robots communicate and collaborate	Programming robots to function without humans	The mechanical design of robots	The field focusing on how humans and robots communicate and collaborate.
158	What does "autonomous navigation" entail for robots?	Manually controlled movement	The ability to navigate and make decisions without human input	Following predefined paths only	Limited movement capabilities	The ability to navigate and make decisions without human input.
159	What is a significant concern regarding AI surveillance?	Increased efficiency	Privacy and civil liberties	Enhanced security	Data storage costs	Privacy and civil liberties.
160	What role does AI play in social media?	Reducing user interaction	Personalized content recommendations	Manual content moderation	Eliminating user-generated content	Personalized content recommendations.
161	What is a primary goal of AI research?	To reduce the need for human intelligence	To create systems that can improve human decision-making and efficiency	To simplify all technology	To limit data usage	To create systems that can improve human decision-making and efficiency
162	What does the term "digital twin" refer to in AI?	A physical duplicate of an object	A digital replica of a physical object or system used for simulations	A backup of digital data	A redundant data model	A digital replica of a physical object or system used for simulations
163	How can AI contribute to enhancing education?	By replacing teachers entirely	By providing personalized learning experiences	By simplifying classroom management	By standardizing all teaching methods	By providing personalized learning experiences
164	What does "smart manufacturing" involve?	Manual assembly lines	The integration of AI and IoT to optimize production processes	Traditional manufacturing techniques	Reducing the use of technology	The integration of AI and IoT to optimize production processes
165	What is "digital ethics" in the context of AI?	The study of data storage techniques	The principles guiding the responsible use of digital technologies	The focus on improving digital marketing	The legal framework for software development	The principles guiding the responsible use of digital technologies
166	What is "algorithm transparency"?	Making algorithms more complex	Ensuring that the workings of algorithms are clear and understandable to users	Reducing the need for documentation	Simplifying data analysis	Ensuring that the workings of algorithms are clear and understandable to users
167	What is a key advantage of neural networks?	They require less data	They model complex patterns	They are easy to interpret	They are low-cost	They model complex patterns
168	Recurrent neural networks are ideal for:	Static data	Sequential data	Image processing	Database queries	Sequential data

169	What does kinematics in robotics study?	Control systems	Forces	Motion without forces	Energy use	Motion without forces
170	Which is NOT part of robot programming?	Control systems	Kinematic modeling	User interfaces	Material selection	Material selection
171	What is a challenge in robot programming?	Strength of robots	Aesthetic design	Efficient decision-making	Battery life	Efficient decision-making
172	A key feature of production systems is:	It can't change rules	Uses rules and a database	Only uses neural networks	Always optimal	Uses rules and a database
173	A problem in AI typically involves:	No constraints	Fixed outcomes	Complexity	Static facts	Complexity
174	What is a disadvantage of hill climbing?	Stuck in local maxima	Needs memory	Is always optimal	Can't apply to complex problems	Stuck in local maxima
175	Means-ends analysis focuses on:	Finding shortcuts	Reducing difference to the goal	Action sequence	Solution efficiency	Reducing difference to the goal
176	What represents knowledge with rules?	Semantic networks	Production systems	Neural networks	Decision trees	Production systems
177	Nonmonotonic logic allows:	Static conclusions	Withdraw conclusions	Fixed rules	No uncertainty	Withdraw conclusions
178	Bayesian networks model:	Deterministic relationships	Probabilistic dependencies	Linear equations	Static data	Probabilistic dependencies
179	Connectionist models benefit by:	High interpretability	Handling noisy data	Simple design	Low cost	Handling noisy data
180	Purpose of robot sensors is:	Increase speed	Collect environment data	Save energy	Optimize programming	Collect environment data
181	Which search is guaranteed to find a solution?	Random search	Depth-first	Breadth-first	Hill climbing	Breadth-first

182	Heuristic functions aim to:	Guarantee solutions	Estimate cost to goal	Solve without search	Validate solutions	Estimate cost to goal
183	What is the goal of semantic networks?	Visualize trees	Represent knowledge in graphs	Store data efficiently	Analyze statistics	Represent knowledge in graphs
184	Default reasoning assumes:	Proven true conclusions	Fixed facts	Assumptions from typical cases	Avoids uncertainty	Assumptions from typical cases
185	Prior probability in Bayesian inference means:	After observing evidence	Initial likelihood estimate	Average of outcomes	Maximum likelihood	Initial likelihood estimate
186	In feedforward neural networks, info:	Moves in one direction	Cycles in network	Moves forward/back	Needs feedback	Moves in one direction
187	An example of narrow AI is:	Self-learning robot	AI playing chess	General problem solver	Human-like conversation	AI playing chess
188	Artificial General Intelligence aims to:	Outperform humans	Learn any intellectual task	Automate jobs	Process large datasets	Learn any intellectual task
189	AI enhances user experience in:	Manufacturing	Customer service	Accounting	Transportation	Customer service
190	Computer vision refers to:	Interpret visual info	Design screens	Simulate human vision	Create graphic designs	Interpret visual info
191	In search algorithms, optimality means:	Speed	Best solution cost-wise	Simplicity	Number of solutions	Best solution cost-wise
192	Exhaustive search involves:	Searching a subset	Searching all solutions	Using heuristics	Prioritizing paths	Searching all solutions
193	Solution space refers to:	Problem definitions	All solutions to a problem	Robot environment	Database limits	All solutions to a problem
194	Which search algorithm doesn't guarantee solutions?	Depth-first	Breadth-first	A* search	Uniform-cost search	Depth-first

195	Frames represent knowledge in:	Logic programming	Semantic networks	Neural networks	Decision trees	Semantic networks
196	What does the frame problem deal with?	Representing changing knowledge	Database limits	Interface issues	Real-time processing	Representing changing knowledge
197	Predicate in logic represents:	Physical action	Condition/relationship	Data structure	Storage	Condition/relationship
198	Certainty factors express:	Absolute truths	Degrees of belief	Statistical data	Model validation	Degrees of belief
199	What does Bayesian reasoning use?	Random sampling	Probabilistic inference	Deterministic algorithms	Static analysis	Probabilistic inference
200	A multi-objective problem involves:	Multiple solutions	Single solution	Random outcomes	Simple structure	Multiple solutions